

FARMDOST AI Farmer assistant

Codec Technologies

A FOUR WEEK INTERNSHIP REPOT

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IILM University
Ai/ML semester 2

Ai/ML intern

July-August 2026

Internship Overview

This internship provided hands-on experience in the field of AI and its implementation in real world solutions , specially in the field of faring and associated fields.

Candidate details

Name - Aman Tiwari

roll no - 25scs1003001515

Degree - B.tech CSE(ai/ml)

Batch - 2025-29

University - IILM university



Department

Artificial intelligence



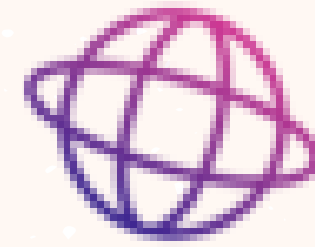
Duration

July-August 2026



Mentor

Abhishek Dubey



CODĒC
TECHNOLOGIES

Company Codec Tech

Ai solution and implementaion driven company. They work on real world projects with real world field work and solution.

Internship Profile

- Program name - Codec ai internship
- Domain - Ai MI and implementation
- Duration - 4 week (1st July - 1st August 2026)

Problem Statement and sol

Farmers specially in our nation are in desperate need for free and reliable Ai solution that can help. Agriculture is a critical global sector, but crop yields and food quality are continuously threatened by plant diseases. Traditionally, identifying these diseases is a time-consuming process that requires specialized expert



Key Responsibilities

- Ai solution
- Deployable adn scalable
- Resource light
- Low cost solution
- Reliable
- Easy to use

Technologies Used

➤ **PYTHON**
CNN (Convolutional Neural Network)
TensorFlow / Keras
Seaborn
Streamlit



Objectives

The main objectives successfully accomplished by this project are:

1. To build a machine learning model capable of accurately detecting and classifying specific plant diseases using leaf images.
2. To preprocess and augment a large collection of leaf images to train an efficient CNN model.
3. To evaluate the model's performance rigorously using standard accuracy and classification metrics.
4. To develop a user-friendly interface for image upload and real-time prediction.
5. To provide a fast and automated solution for plant disease diagnosis in agricultural settings.

Project scope and internship challenges

10. Challenges Faced and Applications

10.1. Challenges Faced

- **Overfitting:** A primary challenge, addressed through extensive **Data Augmentation** and **Dropout** layers.
- **Dataset Imbalance:** Some diseases had fewer examples; handled by class weighting or oversampling.
- **Computational Cost:** High training time for deeper CNN models; addressed by leveraging GPU resources and efficient Transfer Learning models.
- **Real-World Generalization:** Bridging the gap between clean dataset images and blurry/varying real-world field images.

10.2. Applications

- **Farmer Assistance:** Providing instant diagnosis via mobile applications.
- **Precision Agriculture:** Enabling targeted treatment, reducing pesticide use and costs.
- **Automated Monitoring:** Integration into greenhouse systems for continuous, automated inspection.
- **Research:** Serving as a standardized tool for rapid disease classification.

11. Future Enhancements

The system is robust but has potential for significant future growth:

1. **Explainable AI (XAI):** Integrate techniques like **Grad-CAM** to generate heatmaps, visually highlighting the infected area the model used for its prediction.
 2. **Mobile Deployment:** Deploy the model on low-power devices using optimized formats like **TensorFlow Lite**.
 3. **Wider Scope:** Expand the dataset to support a broader range of plant species.
 4. **Segmentation:** Implement an object detection model to **segment** (outline) the exact infected area on the leaf.
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Week 1:

System Design and workflow setup

4. Methodology: System Architecture

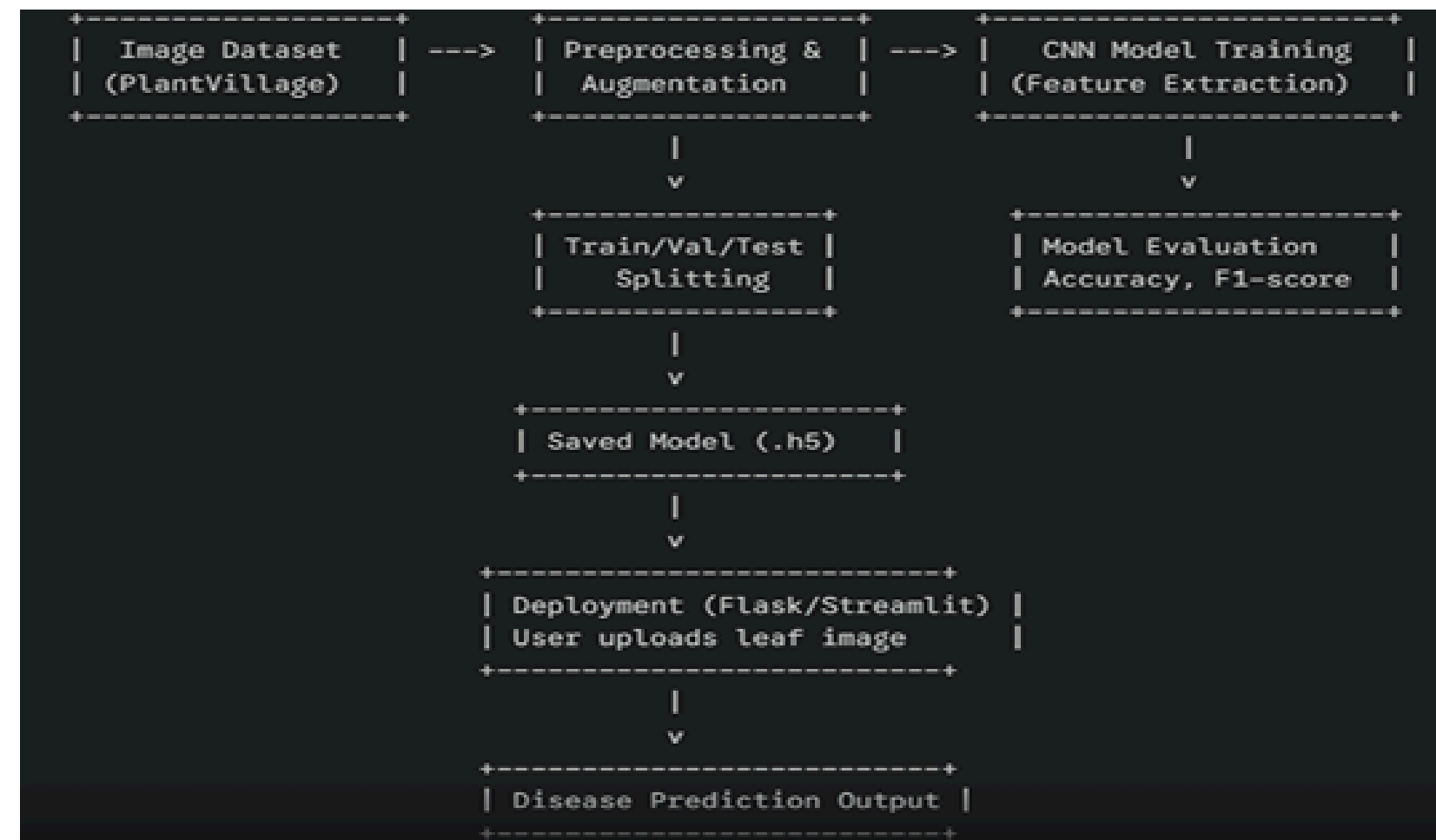
The project employs a standard machine learning pipeline, moving from data preparation to real-time deployment.

4.1. Overall System Workflow

The user-facing system operates through these steps:

1. User uploads a leaf image.
2. Image is **preprocessed** (resized, normalized).
3. Image is passed through the trained **CNN model**.
4. Model outputs the predicted disease class and its **confidence probability**.
5. The final result is displayed to the user.

4.2. Overall Pipeline Diagram



5. Data Preprocessing and Splitting

Robust data preparation is crucial for model performance and stability.

Week 2: Model Selection, Training and Evaluation

5.1. Preprocessing Steps

- **Resizing:** All images are uniformly resized to **128 X 128 pixels**, a common input size for CNN architectures. This smaller dimension helps reduce computational load and training time.
- **Normalization:** Pixel values (0–255) are converted to the range **[0, 1]** to ensure stable and faster training.
- **Data Augmentation:** To prevent **overfitting** and increase the model's robustness to real-world variations, augmentation techniques were applied:
 - Rotation
 - Horizontal/Vertical Flip
 - Random Zoom
 - Brightness variation
 - Shift / Shear transforms

Data Preprocessing

Training Image preprocessing

```
In [224]: training_set = tf.keras.utils.image_dataset_from_directory(
    'train',
    labels="inferred",
    label_mode="categorical",
    class_names=None,
    color_mode="rgb",
    batch_size=32,
    image_size=(128, 128),
    shuffle=True,
    seed=None,
    validation_split=None,
    subset=None,
    interpolation="bilinear",
    follow_links=False,
    crop_to_aspect_ratio=False
)
```

Found 70295 files belonging to 38 classes.

5.2. Train–Validation–Test Split

The dataset was stratified to ensure reliable evaluation:

- **70% Training data:** Used to train the model's parameters.
- **15% Validation data:** Used to tune hyperparameters and monitor overfitting during training.
- **15% Testing data:** A completely unseen set used for final, unbiased performance evaluation.

7. Training and Evaluation

7.1. Training Configuration

The model was trained using the following configuration:

Parameter	Value
Epochs	10
Batch Size	32
Learning Rate	Adaptive Scheduler used to reduce Learning Rate (LR) when accuracy plateaus
Technique	Early Stopping implemented to prevent overfitting by monitoring validation loss

7.2. Evaluation Metrics and Results

The model's final performance was assessed using a suite of metrics to ensure comprehensive understanding:

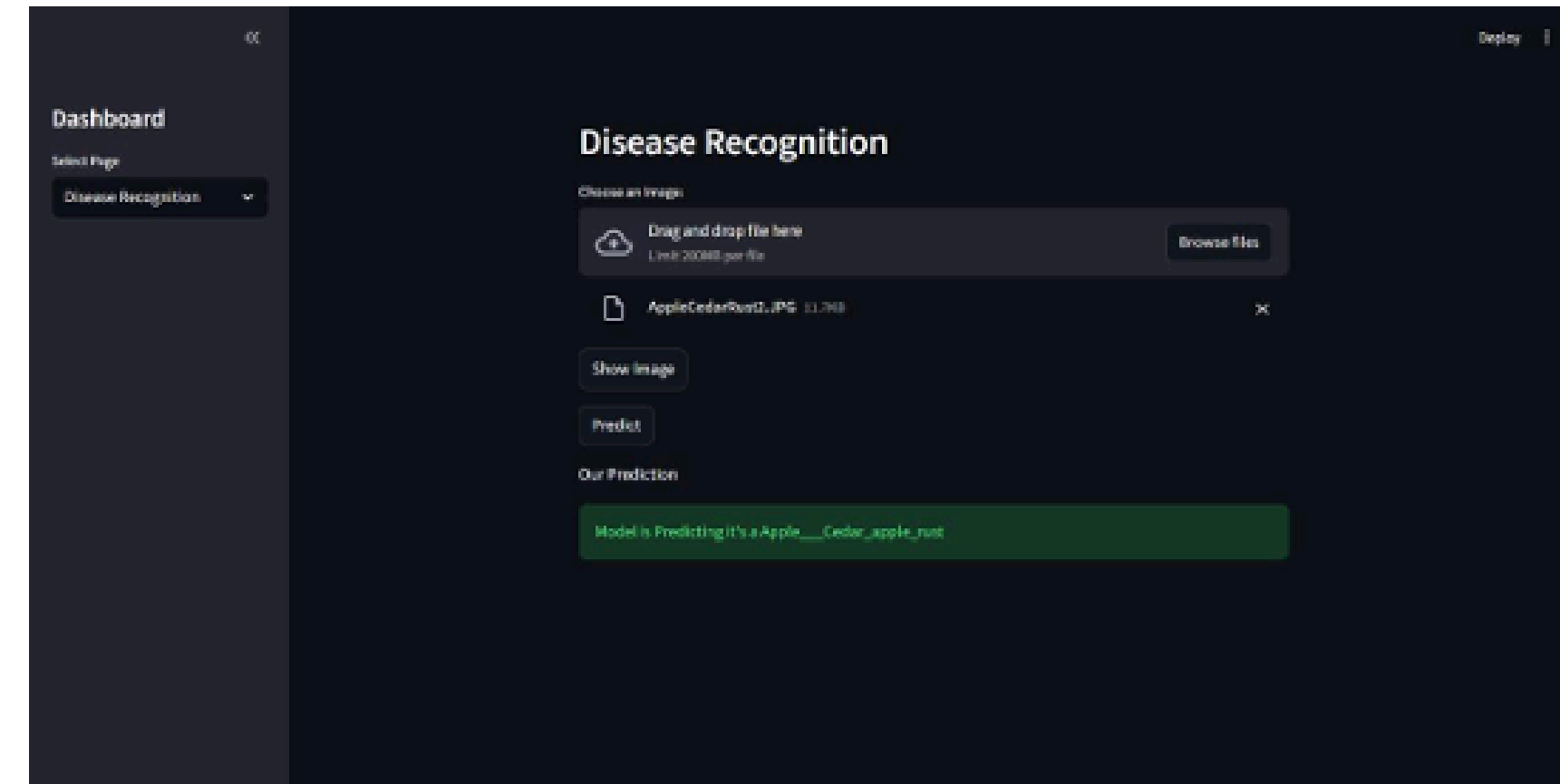
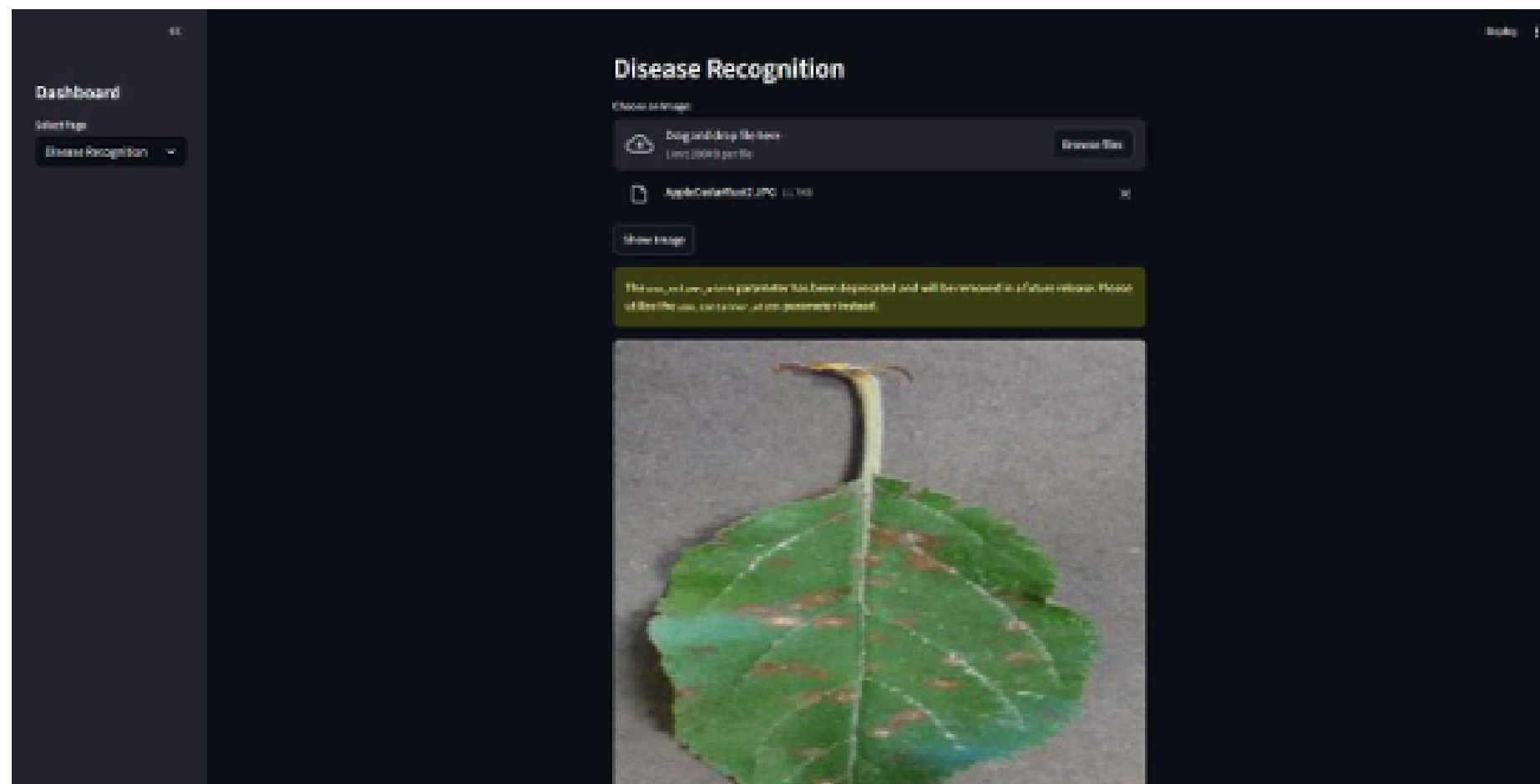
- **Accuracy:** Overall correct predictions.
- **Precision:** The ability of the model not to label as positive a sample that is negative.
- **Recall (Sensitivity):** The ability of the model to find all positive samples.
- **F1-Score:** The harmonic mean of Precision and Recall.
- **Confusion Matrix:** Provides a detailed breakdown of correct and incorrect classifications per class.

Week 3 & 4: Frontend and Deployment

8.1. User Interface Features

The interface, labeled "Disease Recognition," facilitates easy diagnosis:

- **Image Upload:** A simple drag-and-drop or "**Browse files**" area allows the user to upload a leaf image (with a file limit of 200MB per file).
- **Image Visualization:** A "**Show Image**" button displays the uploaded leaf image, confirming the input to the user.
- **Prediction Trigger:** A "**Predict**" button sends the preprocessed image to the trained CNN model running on the backend.
- **Prediction Output:** The result is displayed clearly under "**Our Prediction.**"
 - **Example Output:** The system successfully predicted a sample image, such as `AppleCedarRust2.JPG`, as "**Apple__Cedar_apple_rust**".



This deployment approach makes the complex deep learning model accessible for immediate, real-world application

Key Tech used

9. Technologies & Libraries Used (Revised)

The project leverages powerful tools specifically for model building, visualization, and deployment.

Library / Tool	Category	Purpose
TensorFlow / Keras	Deep Learning	Building, training, and evaluating the Convolutional Neural Network (CNN) model.
Seaborn	Visualization	Plotting graphs and heatmaps, such as the Confusion Matrix, to analyze model flow and metrics.
Streamlit	Deployment	Creating the simple, interactive front-end web interface for real-time image upload and prediction.



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Aman Tiwari

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 01/07/2026 to 01/08/2026



Google for Education
Partner



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, likely belonging to Dr. Anurag Shrivastava.

Dr. Anurag Shrivastava
Program Manager

Thank You

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